

Array Programs

1. Write a program to create static array with 10 elements and display the detail.
2. Write a program to take input of 10 numbers and display numbers.
3. Write a program to create two array with 5 elements (A and B). Perform following operations and display data:
 - a. Add
 - b. Subtract
 - c. Multiply
 - d. Merge
4. Write a program to take input of 10 elements. Take input of single number and find the value from given array and display the position where the value is available in List. Also display error message if the value is not available.
5. Write a program to take input of 10 elements. Display the smallest and largest number from the list.
6. Write a program to perform following data structure operations on array:
 - a. Traversal
 - b. Insertion
 - c. Deletion
 - d. Searching
 - e. Sorting
 - f. Merging
7. Write a program to take input of 3x3 matrix and display data in matrix form.
8. Write a program to take input of 2 (3x3) matrices (A & B). Perform following operations:
 - a. Display both matrices
 - b. Perform A+B and display result
 - c. Perform A-B and display result
 - d. Perform A*B and display result
 - e. Display A-transpose.
9. Write a program to take input of 1D array and display the address of each element. (Use %p to display address as format specifier)

Stack Programs

1. Write a program to perform Push operation on Stack. (Check Overflow condition also)
2. Write a program to perform Pop operation on Stack. (Check Underflow condition also)
3. Write a program to perform Peek (Top) operation on Stack.
4. Write a program to Display all elements of Stack.

5. Write a menu-driven program to implement (Push, Pop, Peek & Display) Stack operations.
6. Write a program to reverse a string using Stack.
7. Write a program to check whether a string is palindrome using Stack.
8. Write a program to check balanced parentheses () in an expression using Stack.
9. Write a program to convert Infix expression to Postfix expression.
10. Write a program to evaluate a Postfix expression using Stack.
11. Write a program to evaluate a Prefix expression using Stack.
12. Write a recursive program to find the Factorial of a number.
13. Write a recursive program to generate the Fibonacci Series.
14. Write a recursive C program to calculate the GCD (Greatest Common Divisor) of two numbers.
15. Write a recursive C program to solve the Tower of Hanoi problem.

Queue Programs

1. Write a program to implement a Simple Queue using array with Enqueue (Insert), Dequeue (Delete) and Display operations. (Check the empty or Full queue whenever require)
2. Write a program to implement Menu-Driven Queue using an Array.
3. Write a program to insert N elements into a queue and display the queue.
4. Write a program to delete N elements from a queue and display the updated queue.
5. Write a program to implement a Circular Queue using array. Perform Insert, Delete and Display operations.
6. Write a program to implement a Double Ended Queue (Deque)
7. Write a program to implement an Input Restricted Deque.
8. Write a program to implement an Output Restricted Deque.
9. Write a program to implement a Priority Queue using an Array.
10. Write a program to insert and delete elements in a Priority Queue according to their priority.